Try yourself

If the atomic number of carbon is 6, what is the number of electrons revolving in its orbit?

4.3.2 Mass number (A) or Atomic mass

We have seen that the mass of an atom is concentrated in its nucleus. From this, we can get the mass number (A). It is equal to the sum of the number of protons (p) and number of neutrons (n) in the nucleus.

Atomic mass or Mass number

= Number of Protons + Number of Neutrons

$A = p + n$

Lithium atom contains 3 protons and 4 neutrons. Its mass number (A) = 3+4 = 7. In a sodium atom, there are 11 Protons and 12 neutrons. Hence, its mass number (A) is 23 (11 + 12).

Try yourself

1. Why the atomic numbers and mass numbers are always whole numbers?
2. A sulphur atom contains 16 protons and 16 neutrons. Calculate its atomic number and mass number.

While writing the symbol of an element, its atomic number and mass number are also written. For example, the symbols of hydrogen, carbon and oxygen are written as $_1H^1$, $_6C^{12}$, $_8O^{16}$ respectively. All the elements in the periodic table have the following combination of protons, electrons and neutrons.

Table 4.2 Symbols of elements

Element	Symbol	Number of proton, electron, neutron,
Carbon	$_6C^{12}$	**6p,6e,6n**
Beryllium	$_4Be^9$	4p,4e,5n
Nitrogen	$_7N^{14}$	7p,7e,7n
Boron	$_5B^{11}$	5p,5e,6n

Isotopes

Atoms of element can have different number of neutrons. Such atoms will have same atomic number but different mass numbers. These atoms are called isotopes. For example, hydrogen has three isotopes. They are: Proteum ($_1H^1$), Deuterium ($_1H^2$), Tritium ($_1H^3$).

Isobars

Atoms that have the same mass number but different atomic numbers are called isobars. Example: Calcium ($_{20}Ca^{40}$), Argon ($_{18}Ar^{40}$).

Table 4.3 Elements and their symbols with their atomic number and mass number

Element	Symbol	Atomic number	Protons (p)	Neutrons(n)	Mass number(p+n)
Hydrogen	H	1	1	0	1
Helium	He	2	2	2	4
Aluminium	Al	13	13	14	27
Oxygen	O	8	8	8	16
Sodium	Na	11	11	12	23

VII Std Science Term-1 EM Unit 4.indd 47 23-01-2025 12.46.22 PM